AMNA ARSHAD

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PROFESSIONAL SUMMARY

A third-year Computer Science student at COMSATS University Islamabad, Wah Campus, with a strong foundation in core technical concepts, full stack software development, and machine learning architectures. Demonstrated experience in building MERN and Python web platforms alongside mobile applications. Proven operational leadership and organizational communication through active involvement in student societies, seeking to leverage academic knowledge and modern engineering skillsets in a professional technical environment.

TECHNICAL SKILLS

Languages: Python, C++, Dart, PHP, HTML/CSS, JavaScript

Frameworks & Libraries: Flutter, Bootstrap 5, Laravel, Scikit-learn, Pandas, NumPy

Databases: SQL, MySQL, MongoDB

Tools & Platforms: GitHub, VS Code, Jupyter Notebook, Google Colab, Tableau, Microsoft Suite

Soft Skills: Leadership, Strategic Planning, Documentation, Team Mentorship, Written and Verbal Communication, Self directed, Attention to detail

EDUCATION

Bachelor of Science in Computer Science

COMSATS University Islamabad, Wah Campus (Expected Graduation: 2027)

Relevant Coursework: Object Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Systems, Web Development, Mobile Application Development, Machine Learning Fundamentals.

PROJECTS

Cybersecurity Analysis Agent | *React.js, Node.js, Express.js, MongoDB, REST API*

- Built a full stack MERN web platform designed for cybersecurity threat monitoring, automated incident response, and live security telemetry analysis.
- Designed a modular and scalable REST API architecture covering attack logs, threat intelligence feeds, IP blacklisting, and alert management.

Hate Speech Detection System | *Python, Flask, Scikit-learn, Pandas, HTML, CSS (Bootstrap)*

- Developed a full stack web application integrating Machine Learning and Natural Language Processing (NLP) pipelines to automate high throughput online content moderation.
- Engineered an end-to-end NLP pipeline utilizing TF-IDF feature extraction techniques.

• Trained, cross-validated, and optimized classification algorithms (SVM, Logistic Regression) to categorize unstructured text data with enhanced F1-scores.

PowerTrack: Electricity Usage & Billing Monitor | *Flutter, Dart, Supabase*

• Conceptualized and designed a comprehensive mobile application to help users monitor real-time electricity consumption and track interactive billing history.

• Implemented modular UI flows and robust state management architectures to ensure a seamless, high-performance user experience for complex meter management.

OPD Connect: Hospital Management System | *HTML, CSS, JavaScript, SQL, PHP*

• Developed a full-stack web application designed to streamline hospital outpatient department operations and scheduling workflows.

• Architected a robust multi-table SQL database schema to scale and handle patient records, automated appointments, and medical histories with strict data integrity.

Predictive Income Modeling | *Python, Scikit-learn, Jupyter Notebook*

• Performed deep Exploratory Data Analysis (EDA) and implemented optimized Linear and Multiple Linear Regression models to accurately predict regional economic indicators.

• Utilized Python data visualization suites to present production trends and optimized mathematical model parameters for precise predictive reporting.

EXPERIENCE

LEADERSHIP & COMMUNITY EXPERIENCE

Vice President - COMSATS Literary Society (Jun 2026 – Present)

- Contributed to strategic planning, administrative management, and constitutional policy updates for society initiatives.

General Secretary - IEEE Robotics and Automation Society (Feb 2026 – Sep 2026)

- Managed official correspondence, participant notifications, and workshop coordination for chapter events.
- Served as an organizer for the online session on AI Prompt Engineering, driving successful event planning and execution alongside organizing multiple online workshops.

General Secretary - COMSATS Literary Society (December 2025 – June 2026)

- Lead 90+ members, coordinate between 6 teams and university administration for event execution
- Manage official society documentation, professional correspondence, and strategic planning for literary initiatives.